

SUNAO SUGIYAMA

List of publications, talks, and press releases, Compiled on August 27, 2026

PUBLICATIONS

The up-to-date list of publication available at [ADS](#).

* = Author list alphabetized

First-author papers or co-authored papers with significant contributions

1. **Sugiyama, Sunao**, R. C. H. Gomes, and M. Jarvis. Fast modeling of the shear three-point correlation function. *Phys. Rev. D*, 113(12):123548, [June 2026:123548](#)
2. **Sugiyama, Sunao**, M. Takada, N. Yasuda, and N. Tominaga. Microlensing constraints on Primordial Black Hole abundance with Subaru Hyper Suprime-Cam observations of Andromeda. *arXiv e-prints*, arXiv:2602.05840, [February 2026:arXiv:2602.05840](#)
3. R. C. H. Gomes, K. Miller, **Sugiyama, Sunao**, et al. Tidal alignment and tidal torquing modeling for the cosmic shear three-point correlation function and mass aperture skewness. *arXiv e-prints*, arXiv:2601.09133, [January 2026:arXiv:2601.09133](#)
4. R. C. H. Gomes, **Sugiyama, S.**, B. Jain, et al. Cosmology with second- and third-order shear statistics for the Dark Energy Survey: Methods and simulated analysis. *Phys. Rev. D*, 112(12):123514, [December 2025:123514](#)
5. **Sugiyama, Sunao** and M. Park. Data compression with noise suppression for inference under noisy covariance. *Phys. Rev. D*, 112(12):123505, [December 2025:123505](#)
6. R. C. H. Gomes, **Sugiyama, S.**, B. Jain, et al. Dark Energy Survey Year 3 Results: Cosmological constraints from second- and third-order shear statistics. *Phys. Rev. D*, 112(12):123515, [December 2025:123515](#)
7. **Sugiyama, Sunao**, R. C. H. Gomes, and B. Jain. Cosmology from a joint analysis of second- and third-order shear statistics with Subaru Hyper Suprime-Cam Year 3 data. *Phys. Rev. D*, 112(12):123545, [December 2025:123545](#)
8. X. Li, T. Zhang, **Sugiyama, Sunao**, et al. Hyper Suprime-Cam Year 3 results: Cosmology from cosmic shear two-point correlation functions. *Phys. Rev. D*, 108(12):123518, [December 2023:123518](#)
9. H. Miyatake, **Sugiyama, Sunao**, M. Takada, et al. Hyper Suprime-Cam Year 3 results: Cosmology from galaxy clustering and weak lensing with HSC and SDSS using the emulator based halo model. *Phys. Rev. D*, 108(12):123517, [December 2023:123517](#)
10. **Sugiyama, Sunao**, H. Miyatake, S. More, et al. Hyper Suprime-Cam Year 3 results: Cosmology from galaxy clustering and weak lensing with HSC and SDSS using the minimal bias model. *Phys. Rev. D*, 108(12):123521, [December 2023:123521](#)
11. R. Dalal, X. Li, A. Nicola, et al. Hyper Suprime-Cam Year 3 results: Cosmology from cosmic shear power spectra. *Phys. Rev. D*, 108(12):123519, [December 2023:123519](#)
12. S. More, **Sugiyama, Sunao**, H. Miyatake, et al. Hyper Suprime-Cam Year 3 results: Measurements of clustering of SDSS-BOSS galaxies, galaxy-galaxy lensing, and cosmic shear. *Phys. Rev. D*, 108(12):123520, [December 2023:123520](#)
13. **Sugiyama, Sunao**, M. Takada, and A. Kusenko. Possible evidence of axion stars in HSC and OGLE microlensing events. *Physics Letters B*, 840:137891, [May 2023:137891](#)
14. H. Miyatake, **Sugiyama, Sunao**, M. Takada, et al. Cosmological inference from an emulator based halo model. II. Joint analysis of galaxy-galaxy weak lensing and galaxy clustering from HSC-Y1 and SDSS. *Phys. Rev. D*, 106(8):083520, [October 2022:083520](#)
15. H. Miyatake, Y. Kobayashi, M. Takada, et al. Cosmological inference from an emulator based halo model. I. Validation tests with HSC and SDSS mock catalogs. *Phys. Rev. D*, 106(8):083519, [October 2022:083519](#)
16. **Sugiyama, Sunao**. Fast Fourier Transformation Based Evaluation of Microlensing Magnification with Extended Source. *ApJ*, 937(2):63, [October 2022:63](#)

17. **Sugiyama, Sunao**, M. Takada, H. Miyatake, et al. HSC Year 1 cosmology results with the minimal bias method: HSC $\times$ BOSS galaxy-galaxy weak lensing and BOSS galaxy clustering. *Phys. Rev. D*, 105(12):123537, [June 2022:123537](#)
18. **Sugiyama, Sunao**, V. Takhistov, E. Vitagliano, et al. Testing stochastic gravitational wave signals from primordial black holes with optical telescopes. *Physics Letters B*, 814:136097, [March 2021:136097](#)
19. *A. Kusenkov, M. Sasaki, **Sugiyama, Sunao**, et al. Exploring Primordial Black Holes from the Multiverse with Optical Telescopes. *Phys. Rev. Lett.*, 125(18):181304, [October 2020:181304](#)
20. **Sugiyama, Sunao**, M. Takada, Y. Kobayashi, et al. Validating a minimal galaxy bias method for cosmological parameter inference using HSC-SDSS mock catalogs. *Phys. Rev. D*, 102(8):083520, [October 2020:083520](#)
21. **Sugiyama, Sunao**, T. Kurita, and M. Takada. On the wave optics effect on primordial black hole constraints from optical microlensing search. *MNRAS*, 493(3):3632–3641, [April 2020:3632–3641](#)
22. H. Niikura, M. Takada, N. Yasuda, et al. Microlensing constraints on primordial black holes with Subaru/HSC Andromeda observations. *Nature Astronomy*, 3:524–534, [April 2019:524–534](#)

co-authored papers

23. T. Zhang, **Sugiyama, Sunao**, S. More, et al. Modeling galaxy clustering and tomographic galaxy-galaxy lensing with HSC Y3 and SDSS data using the point-mass correction model and redshift self-calibration. *Phys. Rev. D*, 113(10):103529, [May 2026:103529](#)
24. T. Zhang, X. Li, **Sugiyama, Sunao**, et al. Cosmology and source redshift constraints from galaxy clustering and tomographic weak lensing with HSC Y3 and SDSS using the point-mass correction model. *Phys. Rev. D*, 113(10):103530, [May 2026:103530](#)
25. D. Rana, S. More, H. Miyatake, et al. Hyper Suprime-Cam Y3 results: Photo-z bias calibration with lensing shear ratios and cosmological constraints from cosmic shear. *Phys. Rev. D*, 113(6):063534, [March 2026:063534](#)
26. R. Terasawa, M. Takada, T. Kurita, and **Sugiyama, Sunao**. Late-time suppression of structure growth as a solution for the S8 tension. *Phys. Rev. D*, 112(8):083556, [October 2025:083556](#)
27. R. Terasawa, M. Takada, **Sugiyama, Sunao**, and T. Kurita. Testing small-scale modifications in the primordial power spectrum with Subaru HSC cosmic shear, primary CMB, and CMB lensing data. *Phys. Rev. D*, 112(4):043514, [August 2025:043514](#)
28. R. Terasawa, X. Li, M. Takada, et al. Exploring the baryonic effect signature in the Hyper Suprime-Cam Year 3 cosmic shear two-point correlations on small scales: The S8 tension remains present. *Phys. Rev. D*, 111(6):063509, [March 2025:063509](#)
29. K.-F. Chen, I.-N. Chiu, M. Oguri, et al. Weak-Lensing Shear-Selected Galaxy Clusters from the Hyper Suprime-Cam Subaru Strategic Program: I. Cluster Catalog, Selection Function and Mass-Observable Relation. *The Open Journal of Astrophysics*, 8:2, [January 2025:2](#)
30. T. Sunayama, H. Miyatake, **Sugiyama, Sunao**, et al. Optical cluster cosmology with SDSS redMaPPer clusters and HSC-Y3 lensing measurements. *Phys. Rev. D*, 110(8):083511, [October 2024:083511](#)
31. I.-N. Chiu, K.-F. Chen, M. Oguri, et al. Weak-Lensing Shear-Selected Galaxy Clusters from the Hyper Suprime-Cam Subaru Strategic Program: II. Cosmological Constraints from the Cluster Abundance. *The Open Journal of Astrophysics*, 7:90, [October 2024:90](#)
32. J. Shi, T. Sunayama, T. Kurita, et al. The intrinsic alignment of galaxy clusters and impact of projection effects. *MNRAS*, 528(2):1487–1499, [February 2024:1487–1499](#)
33. T. Zhang, X. Li, R. Dalal, et al. A general framework for removing point-spread function additive systematics in cosmological weak lensing analysis. *MNRAS*, 525(2):2441–2471, [October 2023:2441–2471](#)
34. Y. Park, T. Sunayama, M. Takada, et al. Cluster cosmology with anisotropic boosts: validation of a novel forward modelling analysis and application on SDSS redMaPPer clusters. *MNRAS*, 518(4):5171–5189, [February 2023:5171–5189](#)

Other Articles

1. S. Sugiyama, M. Takada, and H. Miyatake. Weak lensing cosmology with subaru hsc data. *ASJ*

TALKS

2026

51. **Observational exploration of primordial black holes**, [Progress in Partcles Physics](#), 2026, Aug., *Oral* (Invited Talk)
50. **Probing Primordial Black Hole Dark Matter with Subaru HSC Microlensing Observations of M31**, [IPNS Joint Experimental-Theoretica Cosmology Seminar](#), 2026, Jun., *Oral* (Invited Talk)
49. **Searching for Primordial Black Holes with Microlensing Data from Subaru HSC**, [ASIAA colloquium](#), 2026, Apr., *Oral* (Invited Talk)
48. **Cosmology from two- and three-point cosmic shear statistics**, Cosmology group meeting, 2026, Apr., *Oral*
47. **Observational Search for Primordial Black Holes Using Subaru/HSC Microlensing, Elucidating the Universe’s Creation with Neutrinos based on International Science Collaborations**, 2026, Mar., *Oral* (Invited Talk)

2025

46. **Observational search of PBH**, [Dark matter and black holes](#), 2025, Dec., *Oral* (Invited Talk)
45. **Data Compression with Noise Suppression for Inference under Noisy Covariance**, [Roman F2F meeting](#), 2025, Nov., *Oral*
44. **Cosmology from a joint analysis of second and third order shear statistics**, [KICP seminar](#), 2025, Oct., *Oral*
43. **Cosmology with third-order shear statistics Applications to HSC and DES**, [Beyond-two-point Statistics Meet Survey Systematics](#), 2025, Sep., *Oral* (Invited Talk)
42. **Probing Physics Beyond LambdaCDM: Precision Cosmology with Gaussian and Non-Gaussian Information from Subaru Data**, [Earth and Space Science Seminar](#), 2025, Jul., *Oral*

2024

41. **Exploring Primordial Black Hole with Microlensing Data: Updates on Analysis Pipeline**, UPenn CfPC workshop, 2024, Nov., *Oral*
40. **Exploring Primordial Black Hole with Microlensing Data: Updates on Analysis Pipeline**, [Focus week on primordial black holes 2024](#), 2024, Nov., *Oral* (Invited Talk)
39. **Cosmology with third-order shear statistics**, Roman F2F meeting, 2024, Oct., *Oral*
38. **Exploring Primordial Black Hole with Microlensing Data**, [Pacific conference](#), 2024, Aug., *Oral* (Invited Talk)
37. **Cosmology from Subaru HSC weak lensing Year 3 data**, [Mifa colloquium](#), 2024, May., *Oral* (Invited Talk)
36. **Cosmology from weak lensing three-point correlation function**, astro/cosmo seminar at CMU, 2024, Feb., *Oral*
35. **Cosmology from Subaru HSC weak lensing Year 3 data**, [Subaru Users Meeting FY2023](#), 2024, Jan., *Oral*

2023

34. **HSC Y3 weak lensing cosmology results**, [CosmoPalooza](#), 2023, Oct., *Oral*

33. **Hyper Suprime-Cam Year 3 Results: Cosmology from Weak Lensing with HSC, Windows on the Universe**, 2023, Aug., *Oral (Invited Talk)*
32. **HSC Year 3 Weak Lensing Cosmology Results**, DESI seminar telecon, 2023, Jun., *Oral*
31. **HSC Year 3 Weak Lensing Cosmology Results**, DESC WL telecon, 2023, May., *Oral*
30. **HSC Year 3 Weak Lensing Cosmology Results**, DESC overall telecon, 2023, May., *Oral*
29. **HSC Year 3 Weak Lensing Cosmology Results**, **HSC webinar**, 2023, Apr., *Oral*
28. **HSC Y3 cosmology results**, **CMB x LSS**, 2023, Apr., *Oral (Invited Talk)*
27. **HSC Year 3 Weak Lensing Cosmology Results**, Euclid WLSWG Telecon, 2023, Apr., *Oral*
26. **Collaborative coding: git and github**, **CD3 Opening Symposium**, 2023, Apr., *Oral*
25. **Cosmology analysis with Subaru HSC Y3 data and SDSS data: cosmological parameter inference in Λ CDM model**, **2023 Spring Annual Meeting of ASJ**, 2023, Mar., *Oral*
24. **Cosmology with Subaru HSC weak lensing data**, **2023 GOPIRA Ph.D. thesis**, 2023, Mar., *Oral*

2022

23. **Cosmology analysis with Subaru HSC Y3 data and SDSS data: a joint analysis of cosmic shear + galaxy-galaxy lensing + galaxy clustering**, **2022 Autumn Annual Meeting of ASJ**, 2022, Sep., *Oral*
22. **Revealing the nature of dark matter with gravitational lensing: weak and microlensing**, **Colloquium at Osaka theoretical astrophysics group**, 2022, Jul., *Oral (Invited Talk)*
21. **HSC cosmology: Joint analysis of galaxy-galaxy lensing and clustering from Subaru HSC and SDSS data**, **77th Annual Meeting of JPS**, 2022, Mar., *Oral*
20. **Exploring Primordial black hole with microlensing observation of Andromeda galaxy**, **Subaru Users Meeting 2021**, 2022, Jan., *Oral*

2021

19. **Joint analysis of galaxy-galaxy lensing and clustering at large scales from Subaru HSC and SDSS data**, **34th astro-theory Symposium**, 2021, Dec., *Oral*
18. **Joint analysis of galaxy-galaxy lensing and clustering at large scales from Subaru HSC and SDSS data**, **10th workshop on observational cosmology**, 2021, Nov., *Oral*
17. **Joint analysis of galaxy-galaxy lensing and clustering at large scales from Subaru HSC and SDSS data**, **2021 Autumn Annual Meeting of ASJ**, 2021, Sep., *Oral*
16. **Exploring Dark Matter Candidates with Microlensing**, **KEK theory seminar**, 2021, Apr., *Oral*

2020

15. **Constraining PBH with HSC microlensing**, IPMU phenomenology lunch journal club, 2020, Dec., *Oral*
14. **Testing stochastic gravitational wave signals by PBH microlensing**, **4th KEK-PH + KEK-Cosmo Joint Lectures and Workshop on "Gravitational Wave"**, 2020, Nov., *Oral (Invited Talk)*
13. **Observational constraint on PBH scenarios with HSC microlensing**, **9th workshop on observational cosmology**, 2020, Nov., *Oral*

12. **Developing a method of cosmological parameter inference from galaxy survey data by Subaru/HSC**, [Summer school for young researchers in astronomy/astrophysics](#), 2020, Aug., *Oral*
11. **Validating a minimal galaxy bias method for cosmological parameter inference using HSC-SDSS mock catalog**, Seminar at Daniel Eisenstein group@CfA, 2020, Aug., *Oral*
10. **Validation of PT-based method and cosmological parameter constraint with HSC-Y1 data**, [2020 Spring Annual Meeting of ASJ](#), 2020, Mar.
9. **Constraints on Primordial Black Holes with Microlensing**, Informal seminar at Takahashi and Asada Labs, 2020, Feb., *Oral*
8. **Validation of PT-based method for cosmology analysis with wide field galaxy survey data**, Seminar at astro group of Hirosaki University, 2020, Feb., *Oral*
7. **Constraints on Primordial Black Holes with Microlensing: Wave & Finite Source Effects / PBH from Multiverse**, [Berkeley Week at Kavli IPMU](#), 2020, Jan., *Oral*

2019

6. **Validation of PT-based method for cosmology analysis of wide field galaxy survey data**, [2019 Autumn Annual Meeting of ASJ](#), 2019, Sep., *Oral*
5. **Test and validation of PT-based cosmology : g-g lensing and clustering**, [PT chat](#), 2019, Apr., *Poster*
4. **On the wave effect of PBH microlensing in the observation of the M31 stars**, [2019 Spring Annual Meeting of ASJ](#), 2019, Mar., *Oral*
3. **Wave Effect on PBH Microlensing**, [Accelerating universe in the dark](#), 2019, Mar., *Poster*

2018

2. **Wave effect on PBH micro-lensing and constraint** **Wave effect on PBH micro-lensing and constraint**, [7th workshop on observational cosmology](#), 2018, Dec., *Oral*
1. **Review of new BAO reconstruction method**, [Summer school for young researchers in astronomy/astrophysics](#), 2018, Aug., *Oral*

PRESS RELEASES

[Primordial black holes and the search for dark matter from the multiverse](#), IPMU, 2020 Dec

[How to see the invisible: Using dark matter distribution to test our cosmological model](#), IPMU, 2024 Apr